Date: Day:

(LAB Task 04)

Question No: 03

original heuristic values:-

```
h={
    "A": 5,
    "B": 4,
    "C": 3,
    "D": 2,
    "E": 2,
    "F": 1,
    "G": 0
}
```

chang value:-

```
h={
    "A": 5,
    "B": 4,
    "C": 10, #changed from 3 to 10
    "D": 2,
    "E": 2,
    "F": 1,
    "G": 0
```

Node	True remaining Cost
A	4
B	4
C	1
D	No path
E	No path
F	0
G1	0

Now path is C → F → G1

1 + 0 = 1

⇒ The original heuristic value 3 Modified 10 and True remaining Cost is ⇒ [1]

⇒ 10 is greater than 1 so; this is overestimate.

Original A*:=

Search order:= A, B, C, F, G1, E, D

Goal path:= A → C → F → G1

Goal path = 4

Expanded nodes = 7

The Question result has no change only Algorithm overestimate maybe sometime overestimate result not wrong but A* optimality cost changed.